DISCOVER

DEFINE

DEVELOP

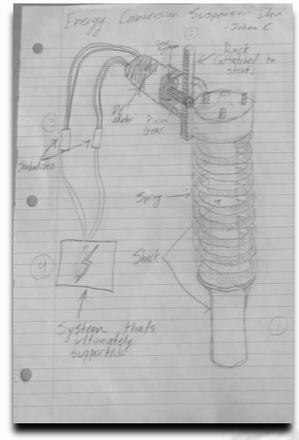
DELIVER

PROBLEM
Suspension problems are difficult to diagnose compared to engine problems. Drivers often hear noises but ignore them because inspection requires going underneath the vehicle.

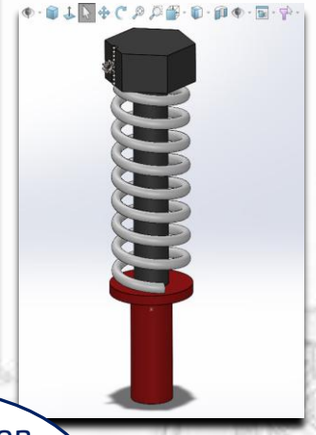
WHAT DRIVERS LOOK FOR
- Reliability
- Comfort
- Low maintenance cost
- Safety

STAKEHOLDERS
- Daily Drivers
- Automotive Technicians
- Vehicle Manufacturers
- Fleet & Commercial Vehicle Owners
- Drivers in Poor Road Conditions

HOW MIGHT WE?
How might we use suspension movement to generate energy while helping drivers become more aware of suspension wear and maintenance needs?



Initial Sketches



- Rack & pinion energy conversion
- Converting vertical motion to rotational
- DC motor



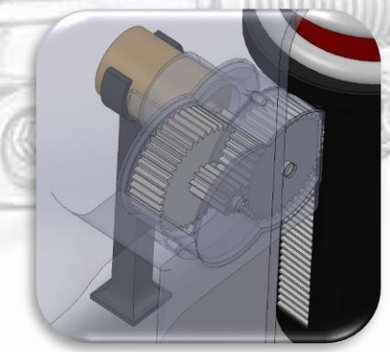
ENERGY CONVERSION SYSTEM
The suspension's vertical movement drives a rack gear connected to a pinion. The rotational motion travels through a gear train to spin a DC motor, generating voltage.



LOW FIDELITY PROTOTYPING
The team created multiple low fidelity prototypes using 3D printed gears , racks, and mounts.

Early testing helped determine proper alignment, gear meshing, and structural stability for the system.

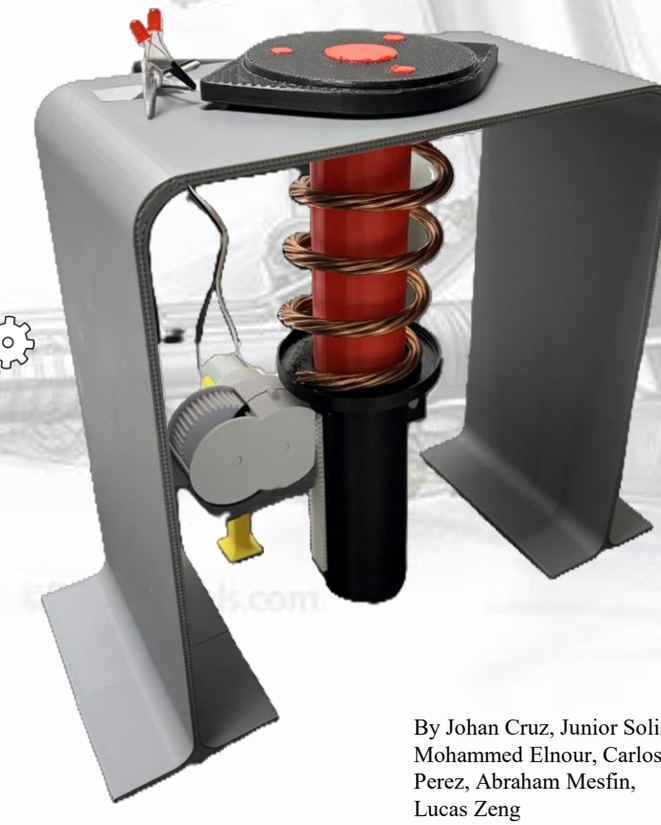




GEAR TRAIN
A compound gear train was designed to increase rotational speed before reaching the generator motor. Faster motor rotation increased voltage output during testing.



RESULTS
- Successful energy generation
- LED illumination achieved
- Stable gear meshing improved performance
- Demonstrated potential for suspension monitoring applications



By Johan Cruz, Junior Solis, Mohammed Elnour, Carlos Perez, Abraham Mesfin, Lucas Zeng